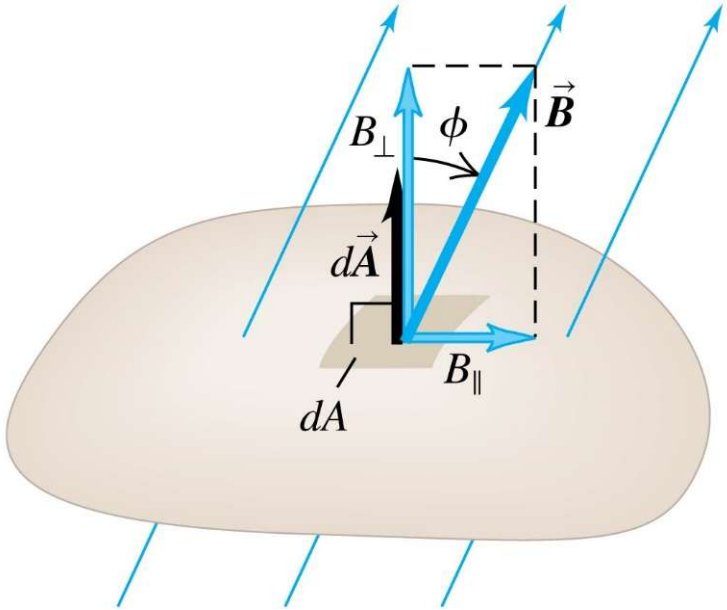
Electromagnetic Induction

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Magnetic Flux

- To define the **magnetic flux**, we can divide any surface into elements of area *dA*.
- The magnetic flux through the area element is defined to be $d\Phi_B = B_\perp dA$.

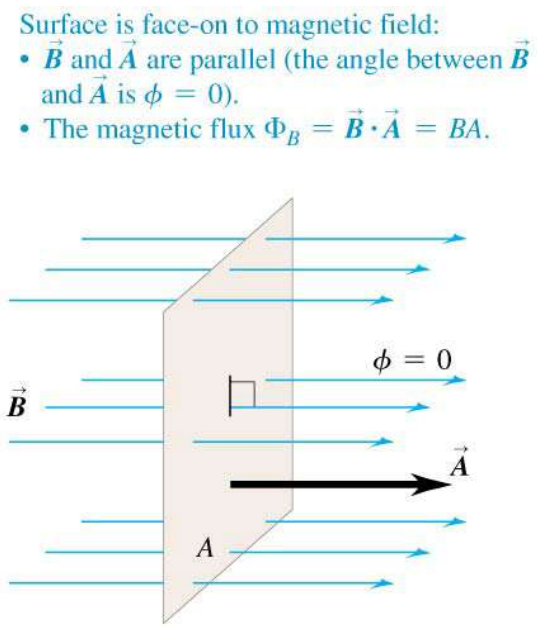


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Magnetic Flux Through a Flat Area: Orientation 1 of 3

- The maximum magnetic flux through a surface occurs when the surface is face-on to the magnetic field.
- In this case the magnetic flux is simply *BA*.



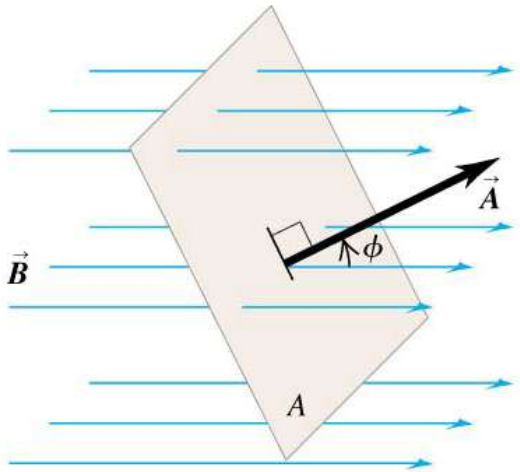
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Magnetic Flux Through a Flat Area: Orientation 2 of 3

- When the surface is at some angle relative to the magnetic field, the magnetic flux is between 0 and *BA*.

Surface is tilted from a face-on orientation by an angle ϕ:
- The angle between $\vec{B}$ and $\vec{A}$ is ϕ.
- The magnetic flux $\Phi_B = \vec{B} \cdot \vec{A} = BA\cos\phi$.



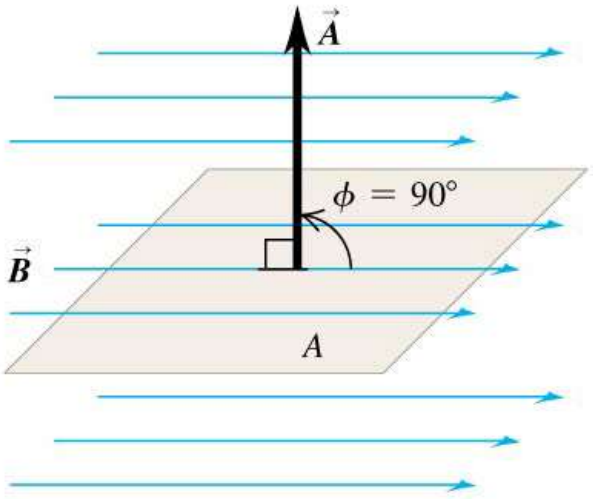
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Magnetic Flux Through a Flat Area: Orientation 3 of 3

- When the surface is edge-on to the magnetic field, the magnetic flux through the surface is zero.

Surface is edge-on to magnetic field:
- $\vec{B}$ and $\vec{A}$ are perpendicular (the angle between $\vec{B}$ and $\vec{A}$ is $\phi = 90°$).
- The magnetic flux $\Phi_B = \vec{B} \cdot \vec{A} = BA\cos 90° = 0$.



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Learning Outcomes

In this chapter, you'll learn…

- how Faraday's law relates the induced emf in a loop to the change in magnetic flux through the loop.
- how to determine the direction of an induced emf.
- how a changing magnetic flux generates a circulating electric field.
- the four fundamental equations that completely describe both electricity and magnetism.
- the remarkable electric and magnetic properties of superconductors.

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Introduction

- The card reader at a gas station scans the information that is coded in a magnetic pattern on the back of your card. Why must you remove the card quickly rather than hold it motionless in the card reader's slot?
- Energy conversion makes use of electromagnetic induction.
- Faraday's law and Lenz's law tell us about induced currents.
- Maxwell's equations describe the behavior of electric and magnetic fields in **any** situation.

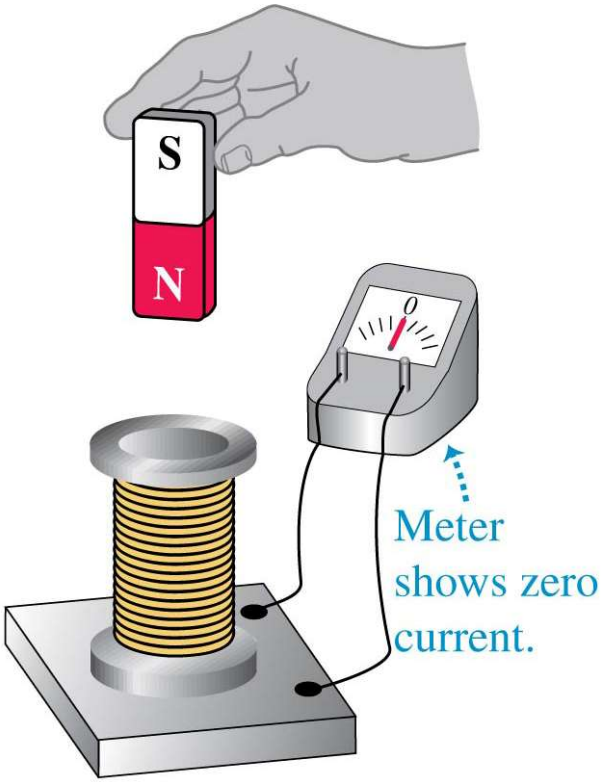


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Induction Experiment (1 of 4)

- During the 1830s, several pioneering experiments with magnetically induced emf were carried out.
- In the figure shown, a coil of wire is connected to a galvanometer.
- When the nearby magnet is stationary, the meter shows no current.

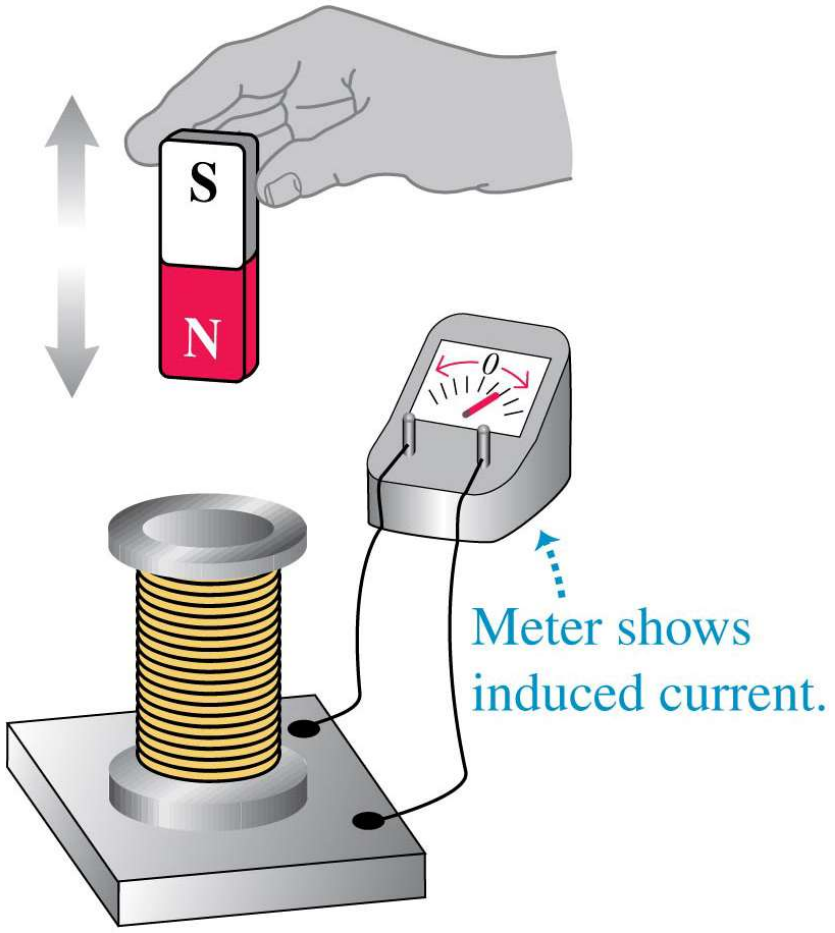


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Induction Experiment (2 of 2)

- When we move the magnet either toward or away from the coil, the meter shows current in the circuit, but only while the magnet is moving.
- We call this an **induced current**, and the corresponding emf required to cause this current is called an **induced emf**.

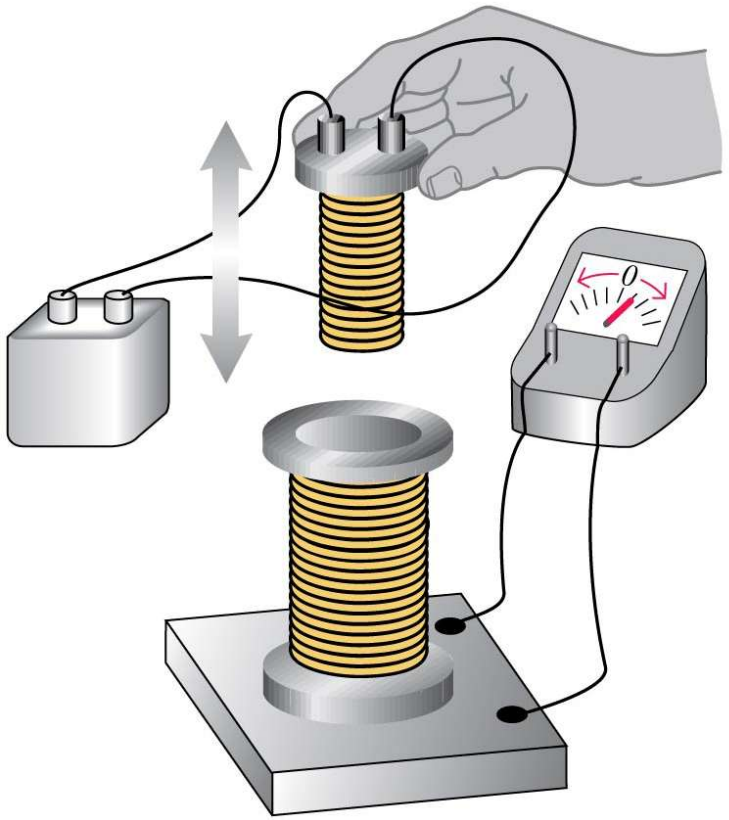


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Induction Experiment (3 of 4)

- In this figure we replace the magnet with a second coil connected to a battery.
- When we move the second coil toward or away from the first, there is current in the first coil, but only while one coil is moving relative to the other.

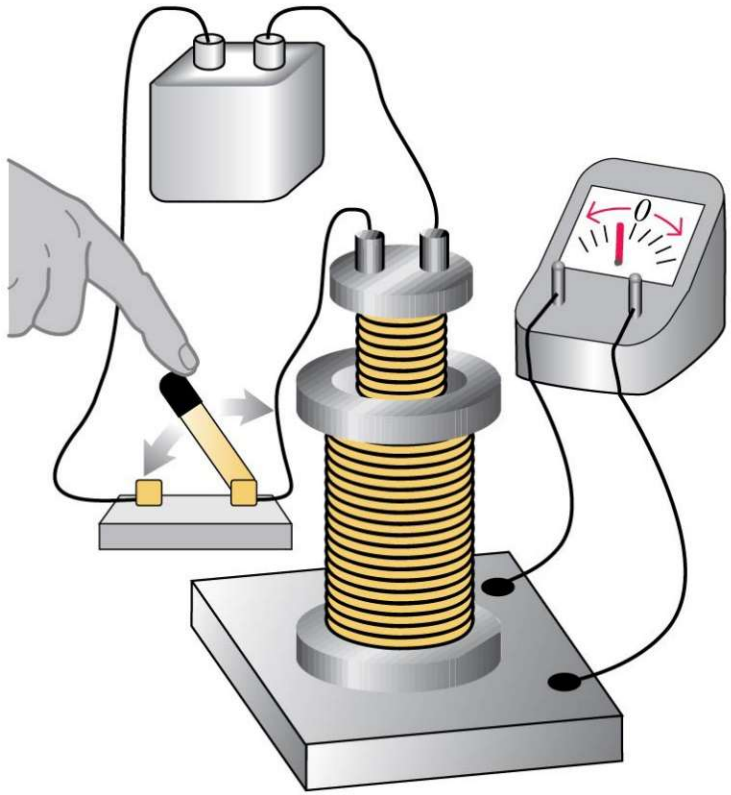


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Induction Experiment (4 of 4)

- Using the two-coil setup of the previous slide, we keep both coils stationary and vary the current in the second coil by opening and closing the switch.
- The induced current in the first coil is present only while the current in the second coil is **changing**.

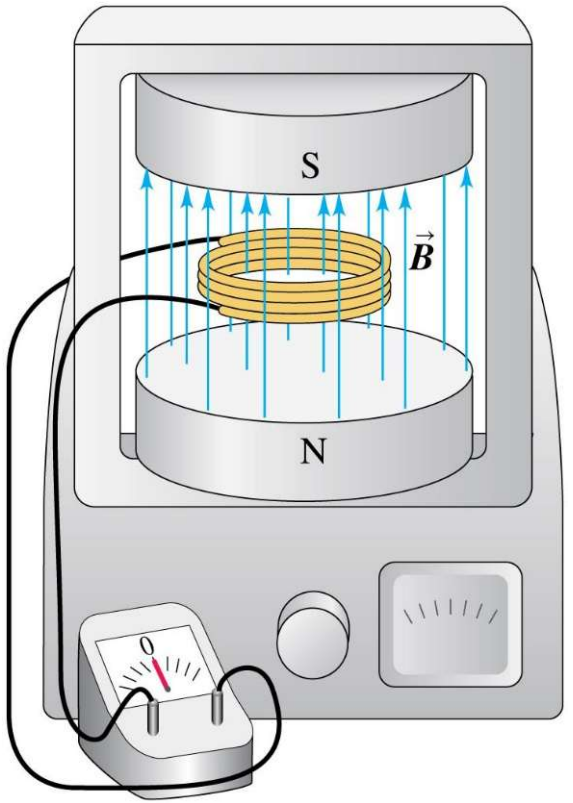


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A Coil in a Magnetic Field

- Shown is a coil in a magnetic field.
- When the magnetic field is constant and the shape, location, and orientation of the coil do not change, no current is induced in the coil.
- A current is induced when any of these factors **change**.



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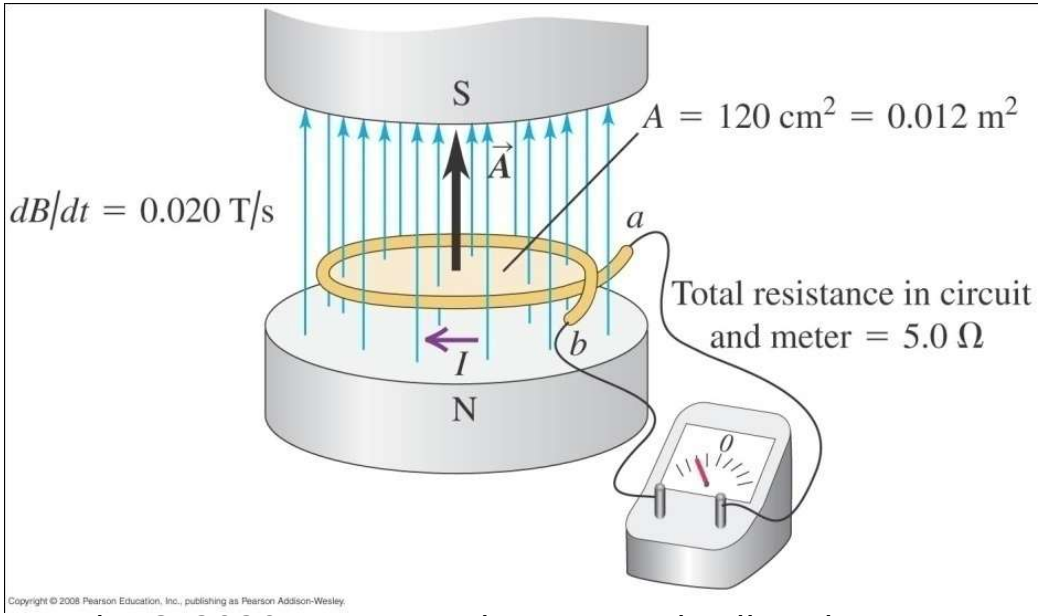
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Example 1

The magnetic field between the poles of the electromagnet in the figure below is uniform at any time but its magnitude is increasing at 0.020 T/s. The area of the conducting loop in the field is 120 cm^2 and the total circuit resistance including the meter is 5.0 Ω.

a) Find the induced emf and the induced current in the circuit.

b) If the loop is replaced by one made of an insulator, what effect does this have on the induced emf and the induced current?



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Faraday's Law of Induction

- When the magnetic flux through a single closed loop changes with time, there is an induced emf that can drive a current around the loop:

Faraday's law: The induced emf in a closed loop ... $\mathcal{E} = -\frac{d\Phi_B}{dt}$ ... equals the negative of the time rate of change of magnetic flux through the loop.

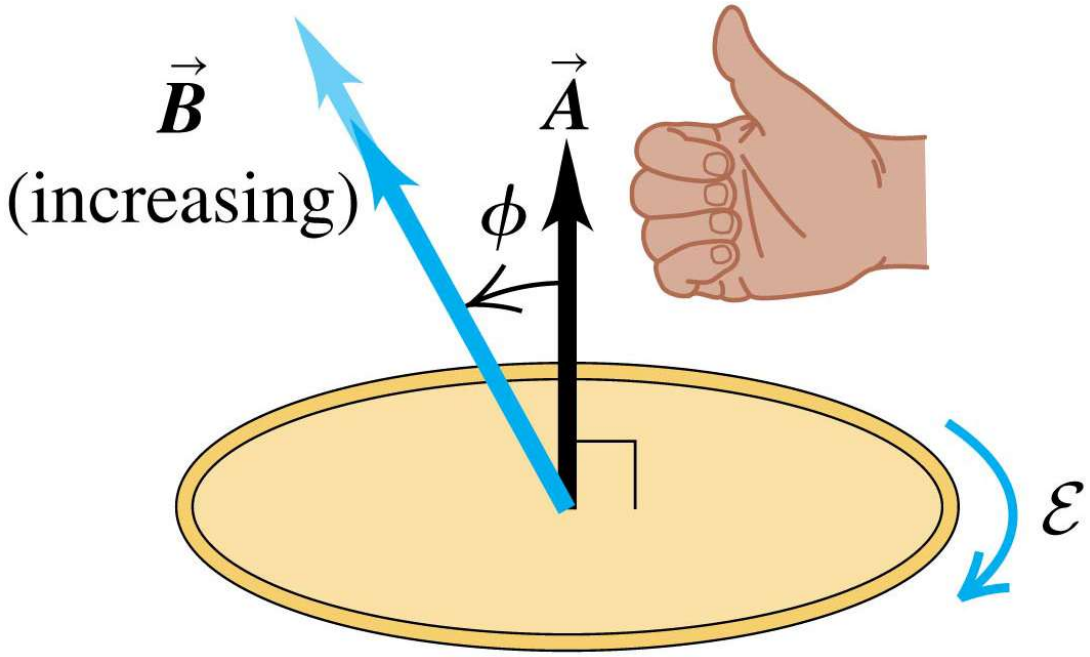
- Recall that the unit of magnetic flux is the weber (Wb).
- $1\ \text{T.m}^2 = 1\ \text{Wb}$, so $1\ \text{V} = 1\ \text{Wb/s}$.

- Video Tutor Solution: Example 29.1

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Determining the Direction of the Induced Emf (1 of 4)

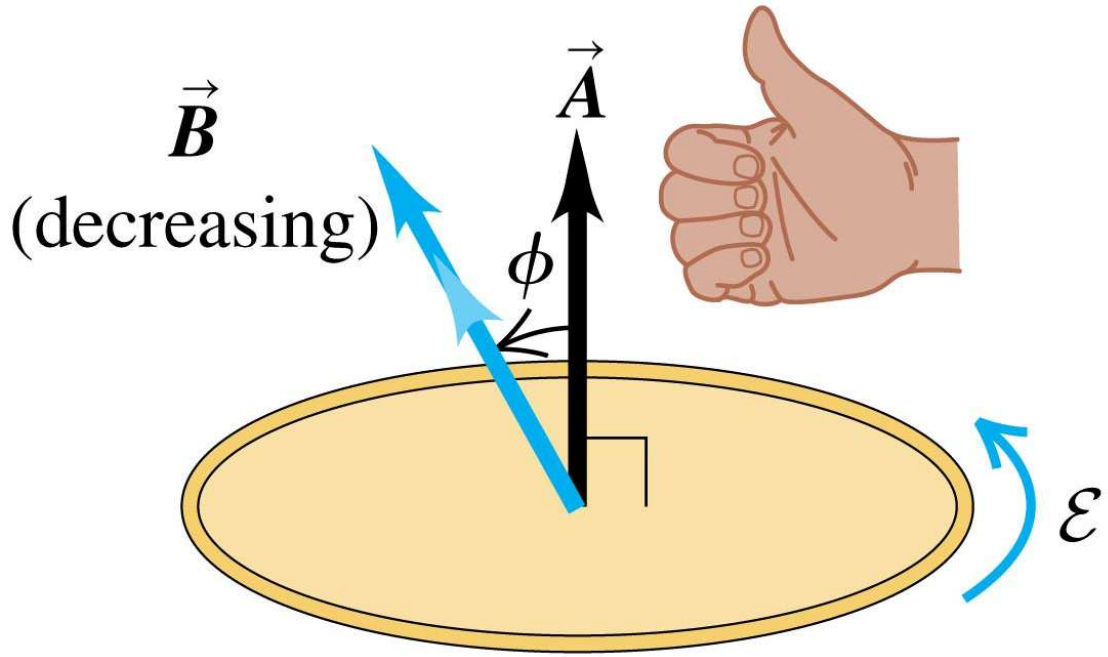


- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

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Determining the Direction of the Induced Emf (2 of 4)

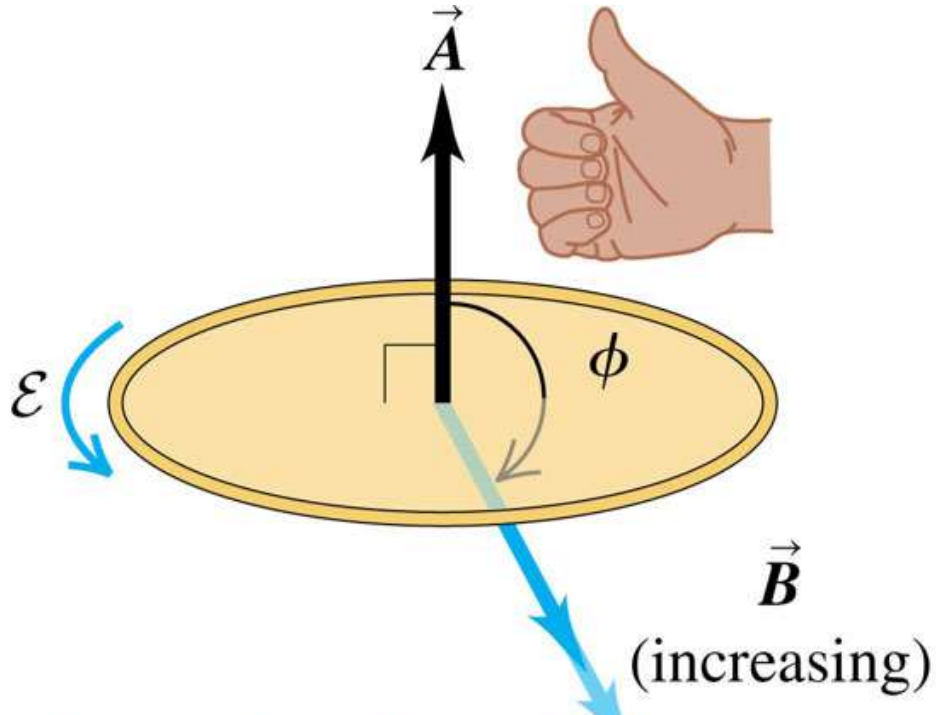


- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).

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Determining the Direction of the Induced Emf (3 of 4)

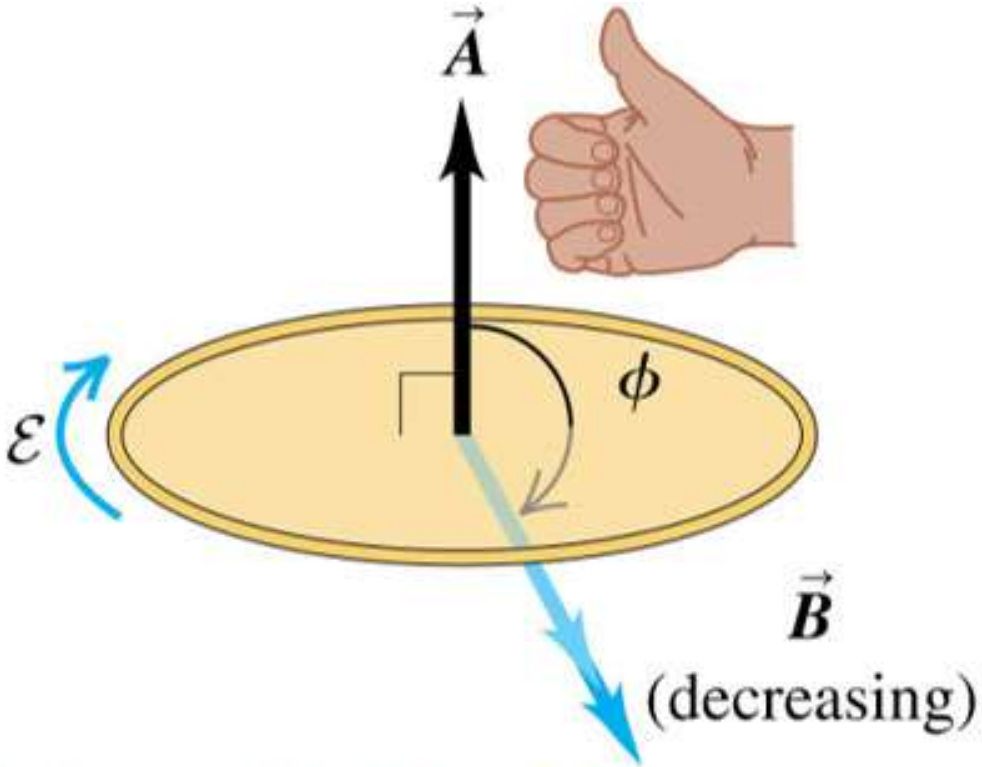


- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).

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Determining the Direction of the Induced Emf (4 of 4)



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

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Faraday's Law for a Coil

- A commercial alternator uses many loops of wire wound around a barrel-like structure called an armature.
- The resulting induced emf is far larger than would be possible with a single loop of wire.
- If a coil has *N* identical turns and if the flux varies at the same rate through each turn, total emf is:



$$\varepsilon = -N\frac{d\Phi_B}{dt}$$

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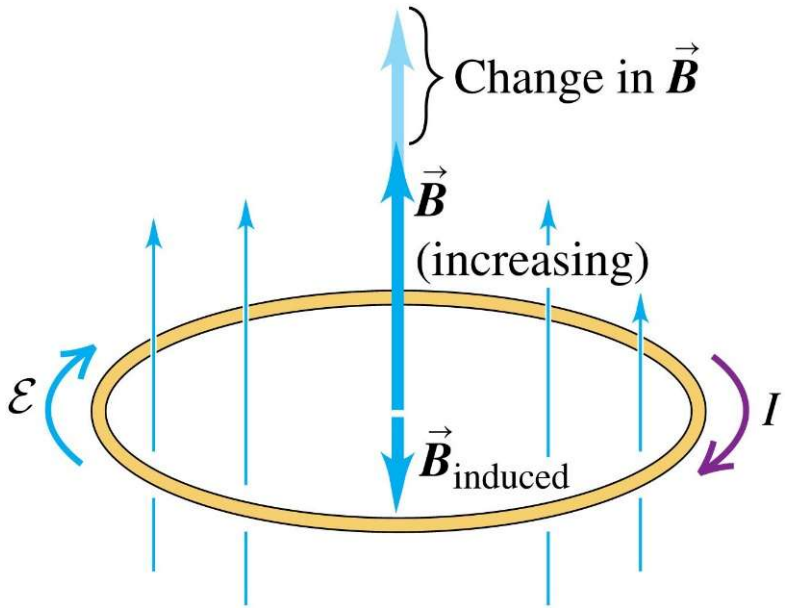
Lenz's Law

- Lenz's law is a convenient method for determining the direction of an induced current or emf:

  The direction of any magnetic induction effect is such as to oppose the cause of the effect.

- For example, in the figure there is a uniform magnetic field through the coil.

- The magnitude of the field is increasing, so there is an induced emf driving a current, as shown.

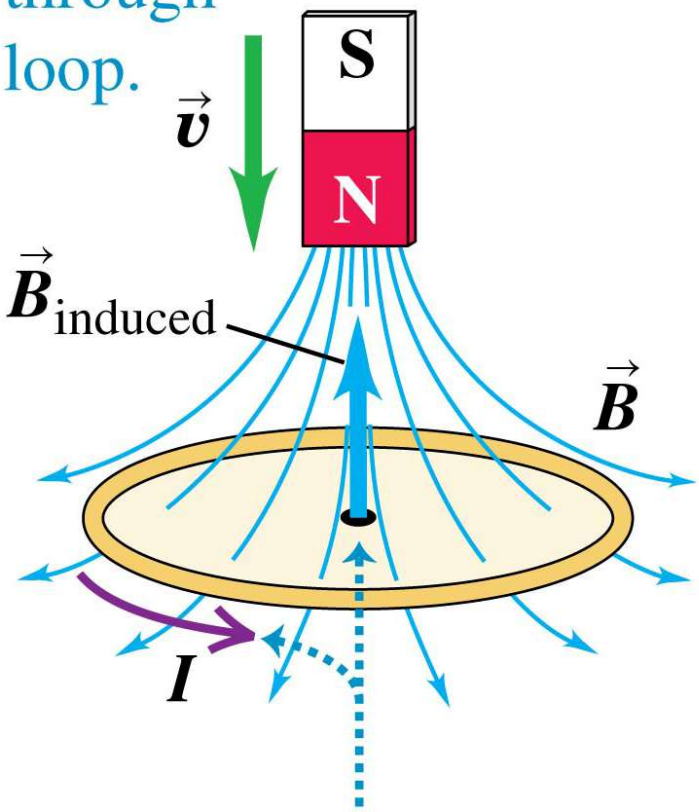


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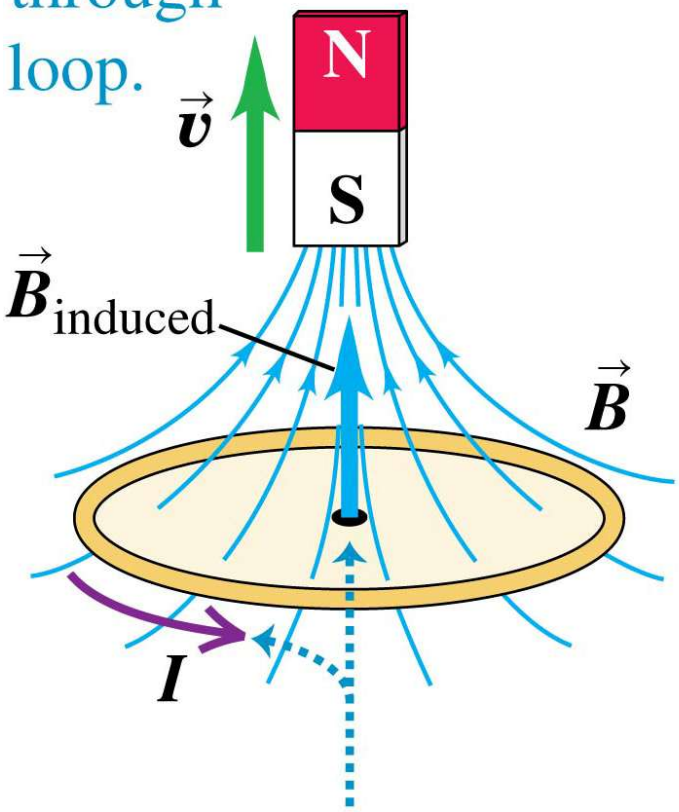
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Lenz's Law and the Direction of Induced Current (1 of 2)

Motion of magnet causes *increasing downward* flux through loop.



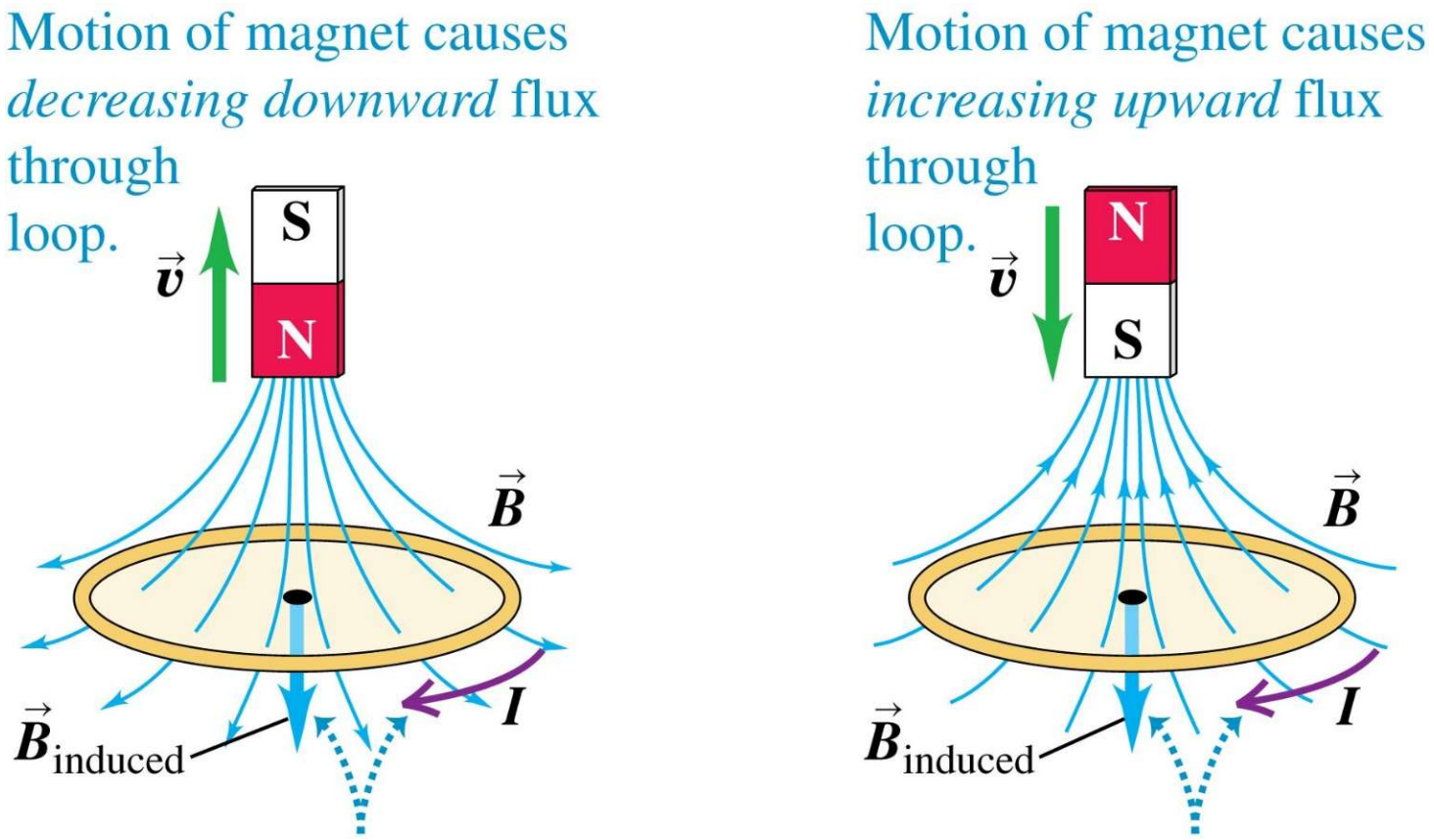
Motion of magnet causes *decreasing upward* flux through loop.



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Lenz's Law and the Direction of Induced Current (2 of 2)



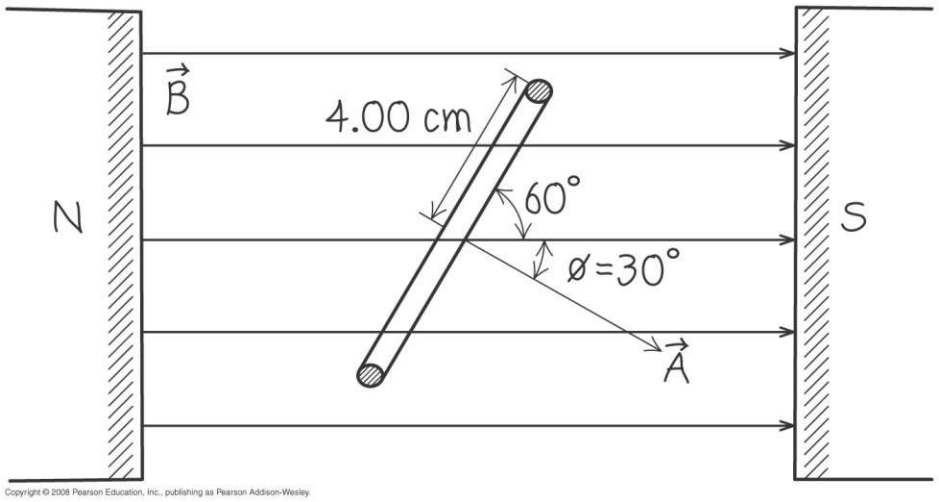
- Video Tutor Solution: Example 29.8

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Example 2

A coil of wire containing 500 circular loops with radius 4.00 cm is placed between the poles of a large electromagnet where the magnetic field is uniform and at an angle of 60° with the plane of the coil. The field decreases at a rate of 0.200 T/s. What are the magnitude and direction of the induced emf?



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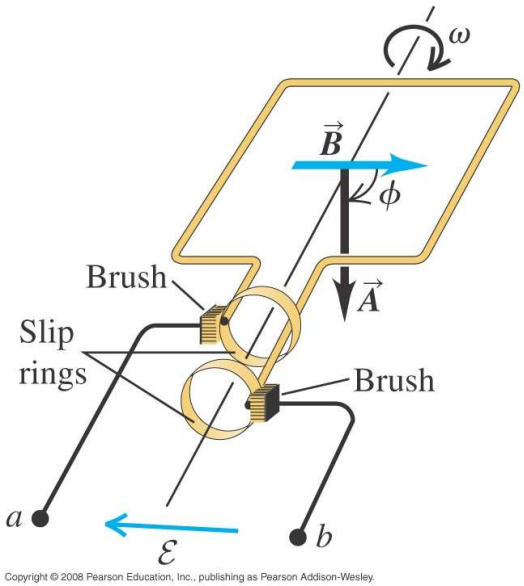
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A simple alternator (a.c. generator)

- A simple alternator consists of a rectangular loop rotating about an axis with constant angular speed ω.

- An emf is produced because the magnetic flux through the loop **changes** as the loop rotates.

- As the loop rotates, the angle ϕ between $\vec{A}$ and $\vec{B}$ increases at a constant rate ω.



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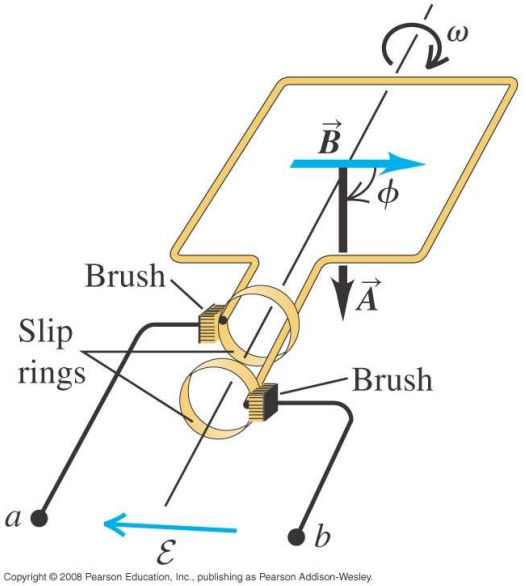
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A simple alternator - cont

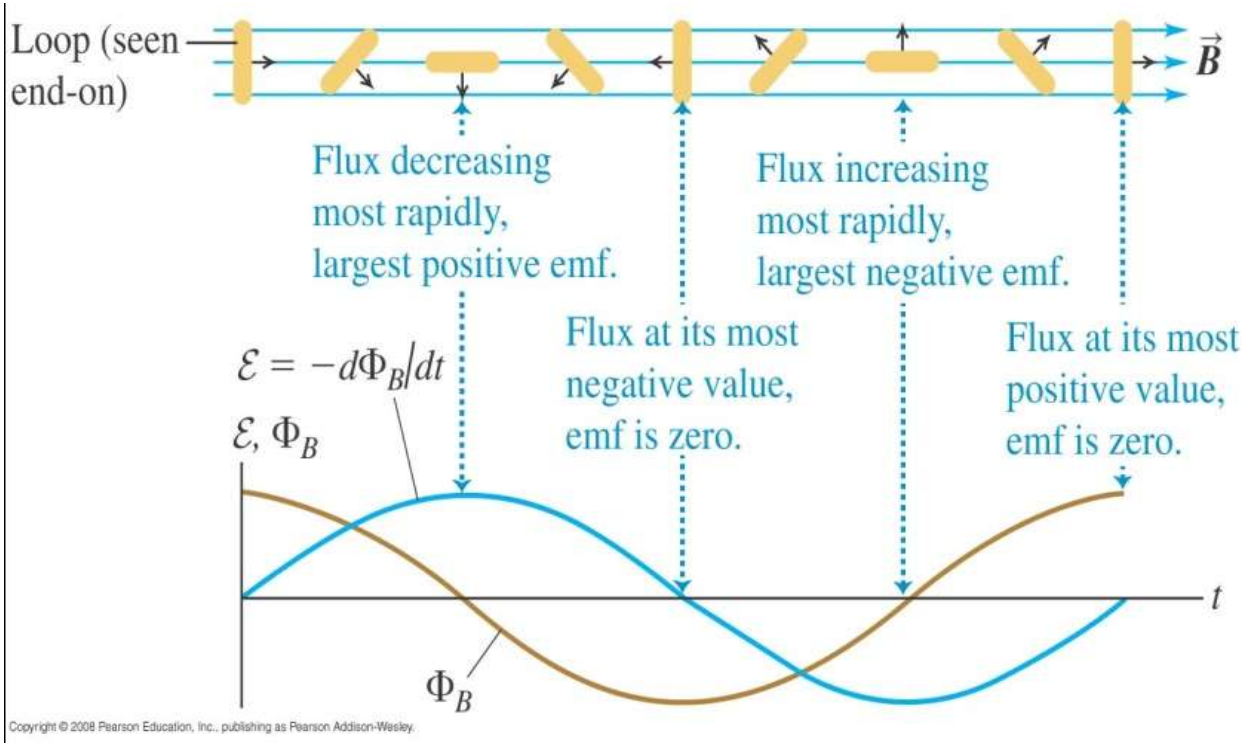
- $$\Phi_B = BA\cos(\omega t)$$

$$\varepsilon = -d\Phi_B / dt = \omega BA\sin(\omega t)$$



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- The absolute value of the induced emf is a maximum when Φ_B is zero and zero when the absolute value of Φ_B is a maximum.



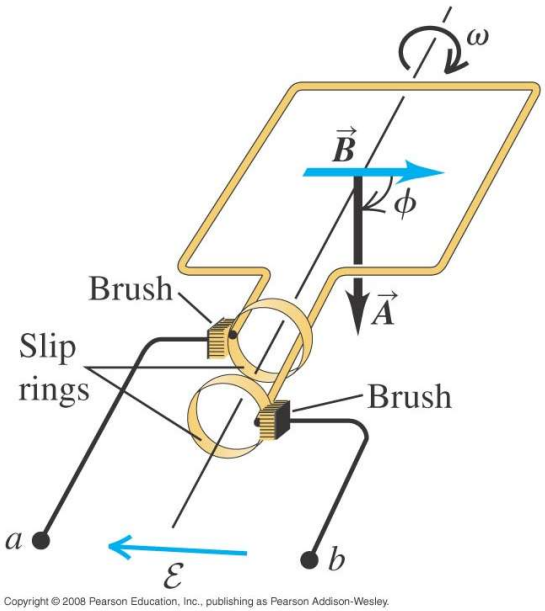
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A simple alternator - cont

- The emf is supplied to the external circuit via two slip rings.
- The amplitude of the emf can be increased by increasing the rotation speed, the field magnitude or the loop area or by using more loops.
- One use of alternators is in automobiles to generate currents in the ignition and the lights.



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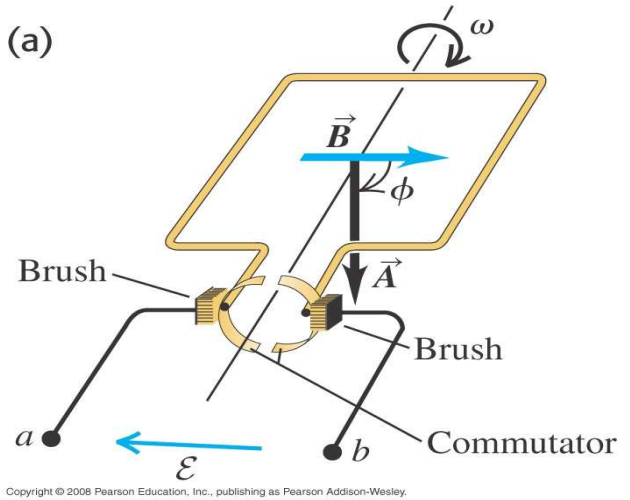
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A direct current (d.c) generator

- A direct current (dc) generator produces an emf that has the same sign throughout the rotation of the coil.

- The working principle is identical to that of an alternator except that a split ring (commutator) is used instead two slip rings.

- The split ring reverses the connections to the external circuit at angular positions where the emf reverses, thus producing an emf always in the same direction.



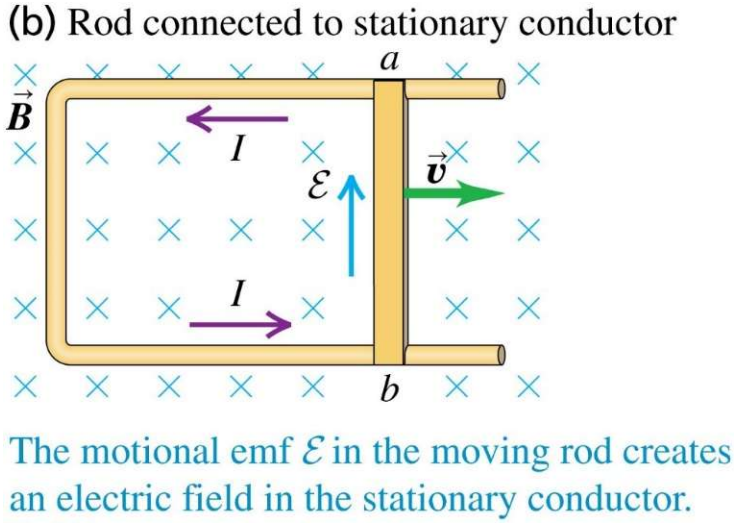
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Motional Electromotive Force

- When a conducting rod moves perpendicular to a uniform magnetic field, there is a **motional emf** induced.



The motional emf $\mathcal{E}$ in the moving rod creates an electric field in the stationary conductor.

Motional emf, conductor length and velocity perpendicular to uniform $\vec{B}$

$$\mathcal{E} = vBL$$

v: Conductor speed; L: Conductor length; B: Magnitude of uniform magnetic field

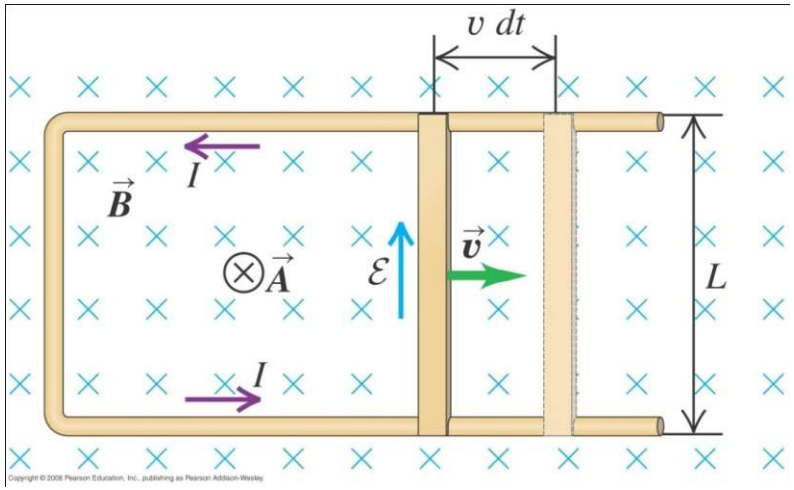
- Video Tutor Solution: Example 29.5

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Example 3

The below figure shows a U-shaped conductor in a uniform magnetic field $\vec{B}$ perpendicular to the plane of the figure directed into the page. A metal rod of length L is laid across the two arms of the conductor forming a circuit. If the rod is moved right with a constant velocity $\vec{v}$, an induced emf and current is set up. Find the magnitude and direction of the resulting induced emf. Note that this set up is also known as slidewire generator.



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Example 4

In the slidewire generator (below figure), energy is dissipated in the circuit owing to its resistance. Let the resistance of the circuit (made up of the moving slidewire and the U-shaped conductor that connects the ends of the slidewire) at a given point in the slidewire's motion be R. Show that the rate at which energy is dissipated in the circuit is exactly equal to the rate at which work must be done to move the rod through the magnetic field.

Force on the rod due to induced current $= iLB$

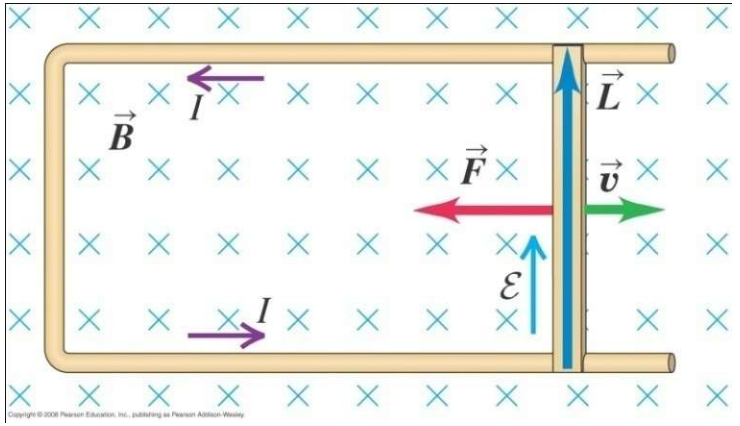
Force is to the left (Fleming 's left hand rule)

Rate of work done to pull the rod to the right $= Fv$

Rate of mechanical work done $= iLBv$

Rate of electrical work done by induced current $= i \times \varepsilon$

$$= i\frac{d}{dt}\Phi = iBLv$$



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Example 5

A conducting rectangular loop of mass M, resistance R, and dimension w by l falls from rest into a magnetic field $\vec{B}$ as shown. During the time interval before the top edge of the loop reaches the field, the loop approaches terminal velocity v_T (terminal velocity occurs when the weight = force on the induced current).

Show that $v_T = \frac{MgR}{B^2w^2}$.

At terminal velocity, $Mg = iwB$

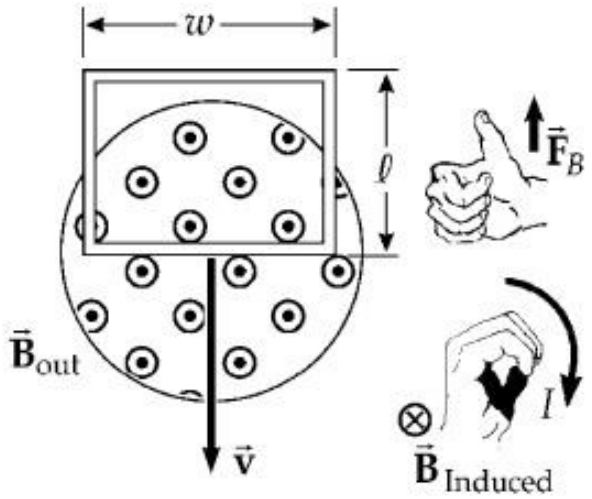
$$i = \frac{\varepsilon}{R} = \frac{1}{R}\frac{d}{dt}\Phi = \frac{1}{R}\frac{d}{dt}(wlB) = \frac{1}{R}Bw\frac{d}{dt}(l)$$

$$= \frac{1}{R}w \times B \times v_T$$

$$= \frac{1}{R}w \times v_T \times B$$

$$Mg = \frac{1}{R}w \times v_T \times B \times w \times B$$

$$v_T = \frac{MgR}{w^2B^2}$$

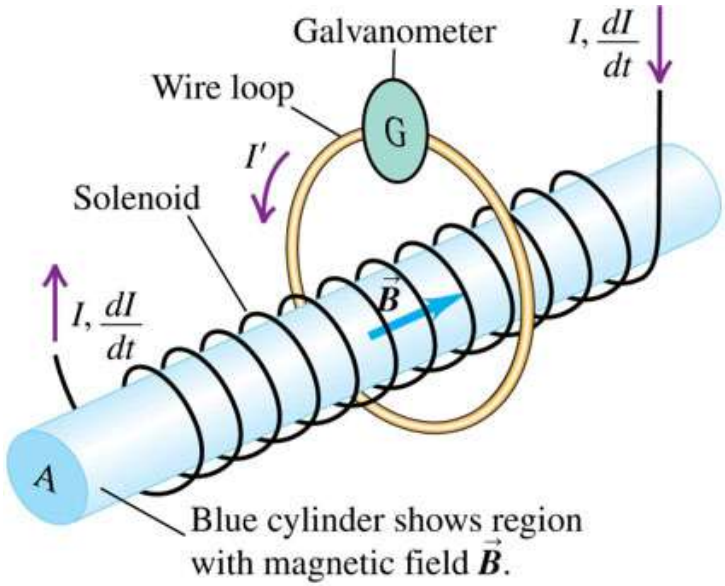


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Induced Electric Fields

- A long, thin solenoid is encircled by a circular conducting loop.
- Electric field in the loop is what must drive the current.
- When the solenoid current *I* changes with time, the magnetic flux also changes, and the induced emf can be written in terms of **induced electric field**:



Faraday's law for a stationary integration path:

Line integral of electric field around path

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

Negative of the time rate of change of magnetic flux through path

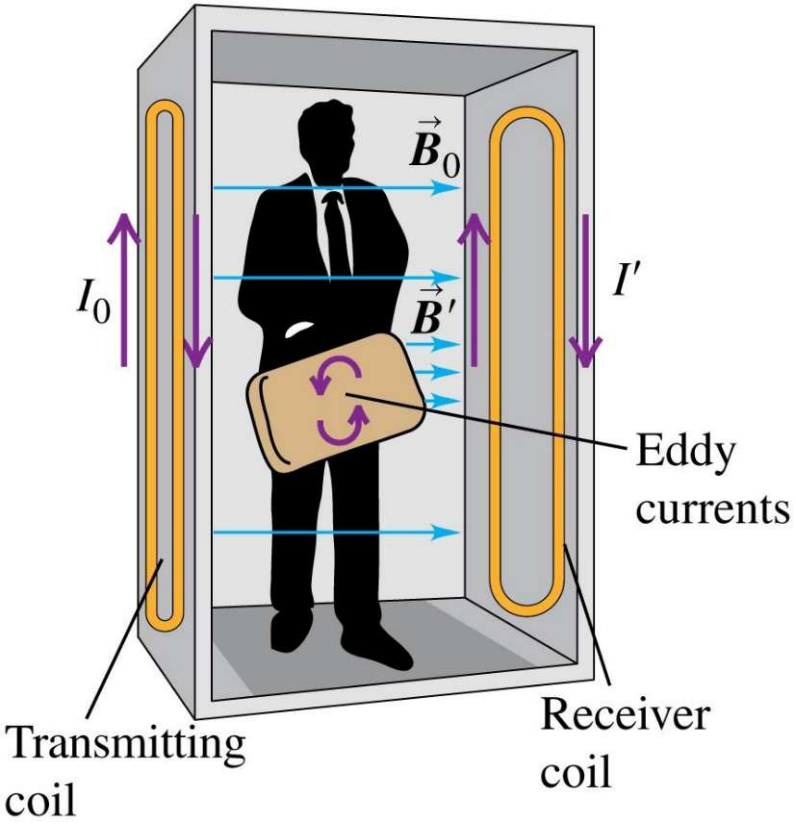
- Video Tutor Solution: Example 29.11

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Eddy Currents

- When a piece of metal moves through a magnetic field or is located in a changing magnetic field, **eddy currents** of electric current are induced.
- The metal detectors used at airport security checkpoints operate by detecting eddy currents induced in metallic objects.



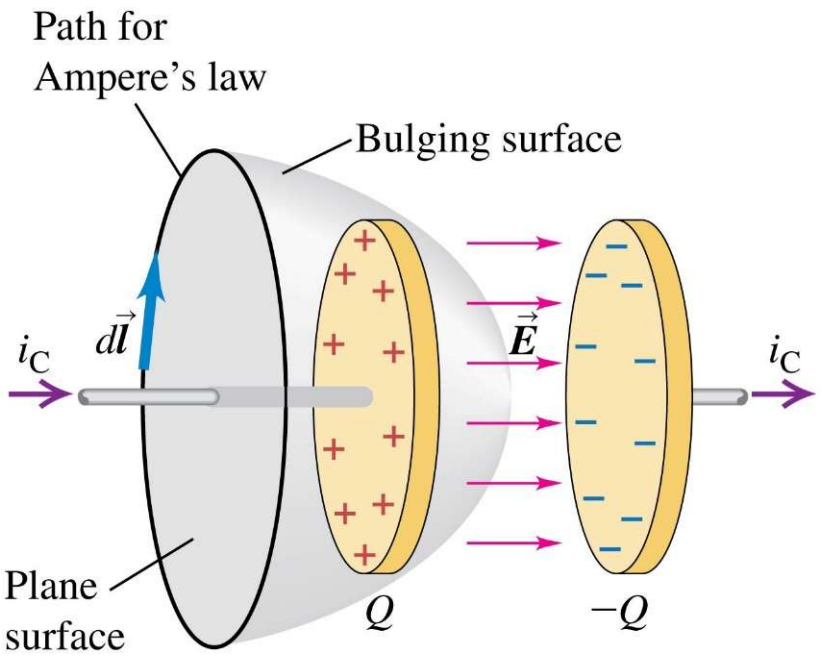
- Video Tutor Demonstration: Eddy Currents in Different Metals

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Displacement Current (1 of 2)

- Ampere's law is **incomplete**, as can be shown by considering the process of charging a capacitor, as shown.
- For the plane circular area bounded by the circle, I_{encl} is the current i_C in the left conductor.
- But the surface that bulges out to the right is bounded by the same circle, and the current through that surface is zero.
- This leads to a contradiction.

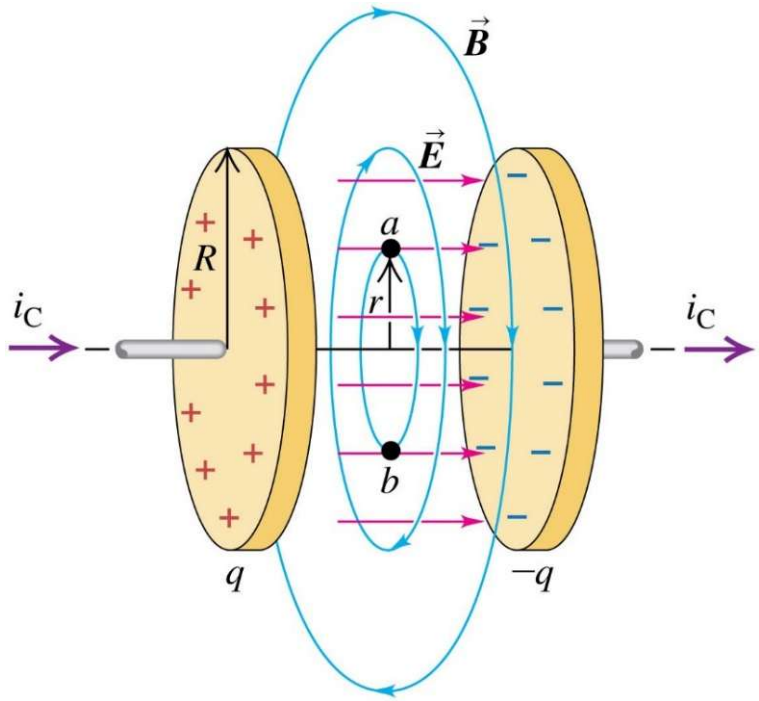


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Displacement Current (2 of 2)

- When a capacitor is charging, the electric field is increasing between the plates.
- We can define a fictitious **displacement current** i_D in the region between the plates.
- This can be regarded as the source of the magnetic field between the plates.



Displacement current through an area $i_D = \epsilon \frac{d\Phi_E}{dt}$ Time rate of change of electric flux through area

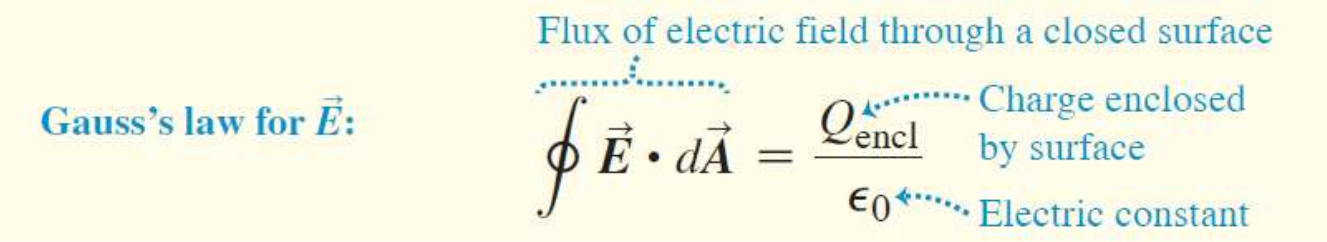
Permittivity of material in area

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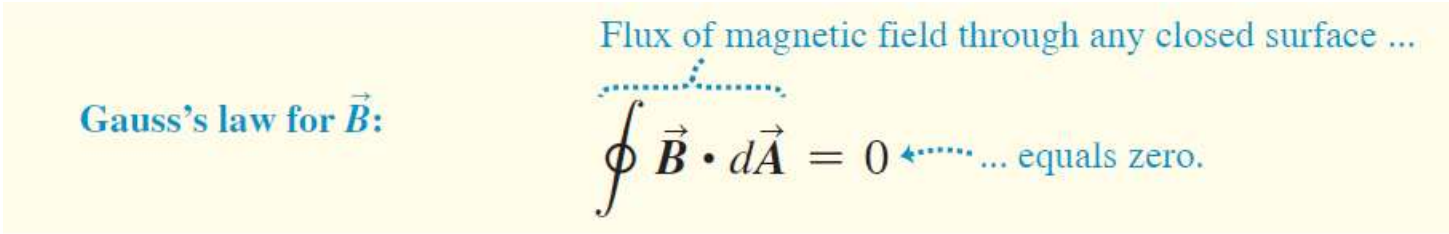
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Maxwell's Equations of Electromagnetism (1 of 2)

- All the relationships between electric and magnetic fields and their sources are summarized by four equations, called **Maxwell's equations**.
- The first Maxwell equation is Gauss's law for electric fields from Chapter 22:



- The second Maxwell equation is Gauss's law for magnetic fields from Chapter 27:



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Maxwell's Equations of Electromagnetism (2 of 2)

- The third Maxwell equation is this chapter's formulation of Faraday's law:

Faraday's law for a stationary integration path:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

Line integral of electric field around path

Negative of the time rate of change of magnetic flux through path

- The fourth Maxwell equation is Ampere's law, including displacement current:

Ampere's law for a stationary integration path:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_C + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}}$$

Line integral of magnetic field around path

Magnetic constant

Conduction current through path

Electric constant

Time rate of change of electric flux through path

Displacement current through path

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Maxwell's Equations in Empty Space

- There is a remarkable symmetry in Maxwell's equations.
- In empty space where there is no charge, the first two equations are identical in form.
- The third equation says that a changing magnetic flux creates an electric field, and the fourth says that a changing electric flux creates a magnetic field.

In empty space there are no charges, so the fluxes of $\vec{E}$ and $\vec{B}$ through any closed surface are equal to zero.

$$\oint \vec{E} \cdot d\vec{A} = 0$$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

In empty space there are no conduction currents, so the line integrals of $\vec{E}$ and $\vec{B}$ around any closed path are related to the rate of change of flux of the other field.

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End of chapter

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